

Hirmay Sandesara

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EDUCATION

- **Technical University of Munich** Munich, Germany
 - *Master of Science in Computational Science and Engineering; Grade: 1.5 (3.65/4)* Oct 2024 – Present
 - [TUM Merit Based Scholarship](#)
 - [Deutschlandstipendium recipient](#)
 - CSE Student Representative
- **Ahmedabad University** Ahmedabad, India
 - *Bachelor of Technology in Computer Science and Engineering; Grade: 1.4 (3.73/4)* Jul 2019 – May 2023
 - [Rank: 2/140](#)
 - [Dean's Recognition List Awardee, Graduated with certificate of highest excellence](#)

EXPERIENCE

- **Ahmedabad University** Ahmedabad, India
 - *Research Assistant* Sep 2023 - May 2024
 - Developed a **multidimensional framework** for **nonlinear optimal control** problems, addressing limitations in existing frameworks through canonical quantization and constraint Hamiltonian derivation.
 - Designed and implemented a modified **Variational Quantum Algorithm** (VQA) tailored to our problem involving broken-PT symmetric systems. Overcame limitations of traditional approaches for exploring complex system dynamics.
 - Implemented mathematical formulations and the modified VQA, using Python libraries such as PennyLane, Qibo, and SciPy, along with Mathematica for generalising functions, symbolic computing, parallelising integrals, optimising solver methods, and solving partial differential equations. Published in the [Quantum Information Processing Journal](#).
- **Quantum Open Source Foundation** Remote
 - *Mentee* Apr 2024 - Aug 2024
 - Mentored by **Ryan Hill**, CTO of qBraid, alongside 3 other mentees. One of approximately forty participants selected out of a cohort of **more than 200** applicants. Focusing on creating a Python framework for Quantum Reservoir Computing (QRC) with customisable models and having integration with QuEra's devices.
- **Santa Fe Institute** Remote
 - *Student Researcher* Jan 2024 - Apr 2024
 - I collaborated on a research project titled, "Darkening Skies," analyzing the impact of COVID-19 on ecosystems and conservation efforts
- **Artificial Brain** Remote
 - *Quantum Computing Intern* Jul 2022 - Dec 2022
 - Co-authored a white paper titled "[Space Sustainability using Quantum Computing](#)," emphasising on leveraging quantum computing techniques such as **QAOAs**, quantum annealing, and optimisation methods to address challenges related to space debris.
 - Additionally, contributed to the formulation of a **quantum scheduling algorithm** for Earth Observation Satellites. Furthermore, collaborated in the development of a [quantum optimisation algorithm](#) aimed to maximise the Sharpe-ratio required for financial portfolio management on D-Wave systems.
- **Mentor** Remote
 - *IBM Quantum Challenge Fall: 2022, 2024* Fall: 2022, 2024
 - Corrected bugs in tasks, assisted participants with challenges and ensured their understanding of relevant topics.

ACHIEVEMENTS

- [2nd Prize, MIT iQuHack IonQ 2026 Challenge](#): I led the team which engineered a heuristic-based parity-check entanglement distillation protocol. Achieved a score of 89 and high-fidelity edge claiming (≥ 0.9) with a minimal single Bell-pair budget.
- [QHack Open Hackathon 2024](#)
 - Won [third prize](#) in Seeing the Future Challenge.
 - Won AWS' "Top 3 Prize" and will receive a mentorship opportunity with Michael Brett, Global Lead for Quantum Technologies Business Development at AWS.
- [Winner, QRISE 2024's Classiq Challenge](#) and received certificate for [advanced proficiency in quantum algorithm design](#).
- [IBM Certified Associate Developer - Quantum Computation using Qiskit v0.2X](#).
- [Qiskit Advocate](#)
- [IBM Quantum Challenge 2020](#) and [2021](#) by successfully completing all challenges and earning Advanced badges.
- [Qiskit Challenge India](#): Demonstrated proficiency by achieving perfect scores in all challenges.

SELECTED PROJECTS

- [‘NISQ algorithm for simulating open-quantum systems’](#) Team - 4, Mar 2024
 - *QHack Open Hackathon 2024* *Remote*
 - I led the team “Quantum.Questers” to develop a Python-based framework for simulating open-quantum systems, specifically catered for NISQ devices. Implemented shallow-depth Parameterized Quantum Circuits (PQCs) inspired by existing Universal Quantum Algorithms to address the absence of existing VQA frameworks for specialized problems.
 - We introduced improvements for robustness and usability, with an emphasis on implementing low-depth PQCs for initial state preparation and time evolution. Conducted extensive testing of the state preparation method using both randomly generated and Alzheimer patient data through parallelized GPU optimization computations, using NVIDIA’s CuQuantum library, and explored the effectiveness of cat qubits (“EMU:15Q:LOGICAL_EARLY”).
- [‘Zero Noise Extrapolation method’](#) Individual, Apr 2024
 - *QOSF Screening Task* *Remote*
 - Implemented the Zero Noise Extrapolation (ZNE) method which is a popular quantum error mitigation technique, through a ‘DepolarizingNoiseModel’ class and functions for scaling noise levels through unitary folding method for Qiskit circuits.
 - Applied various extrapolation methods to extrapolate to zero-noise limit and benchmarked the results.
- [Hamiltonian simulation of coupled classical harmonic oscillators](#) Team - 3, Apr 2024
 - *Classiq QRISE 2024 Challenge* *Remote*
 - I led the team “Classiq Entanglers” in simulating coupled classical oscillators on Classiq’s platform, based on the novel algorithm by Babbush et al., aiming for scalability, we showcased practical demonstration on a 70+ qubit system, to potentially showcase quantum supremacy.
 - Developed and implemented state preparation approaches, Hamiltonian formulation method, and various quantum simulation methodologies for simulation of the Hamiltonian, aiming to maximise fidelity while minimising circuit depth.
- [Open-Source Contributions](#)
 - Worked on a contribution for preparing uniform superposition states based on the Shukla-Vedula (24) algorithm for [Qiskit’s Circuit library](#).
 - Co-authored a subsection on Grover’s algorithm chapter in the [Qiskit Textbook](#).
 - Collaborated with co-author Frank Harkins to fix a bug in [Qiskit Terra](#).
 - Translated over 5,000 words of [Qiskit’s educational material](#) into Hindi.

PROGRAMMING SKILLS

- **Languages:** Python, Mathematica, MATLAB, Julia, C, C++, PostgreSQL, HTML
- **Tools/Libraries:** Qiskit, PennyLane, NumPy, SciPy, Qibo, SymPy, QuTiP, Matplotlib, PyTest, Git, LaTeX, Flask